An Enhanced Multi-Modal Medical Image Fusion Technique using Stationary Wavelet Transform and Method Noise based Wiener Filter

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***Abstract*— Image fusion is the procedure of joining the features and details of input medical images from different modalities like Computed tomography (CT) scan, Magnetic Resonance Imaging (MRI), etc. into one fused image that contains the information of input images. This paper proposes a work (M3IF-SWT) that attempts to build a multi-modal medical image fusion (M3IF) technique that integrates MRI and CT images using Stationary Wavelet Transform (SWT) and wiener filter. M3IF-SWT applies SWT to the input MRI and CT images. The detailed coefficients are fused using maximal operation, and the approximate portions are routed to the wiener filter. The average is taken of wiener filtered approximate part. Later, the outcomes are enhanced by using the wiener filter approach to reduce noise. The post-processing operation is done by applying method noise approach using wiener filter. By doing this, the typical SWT approach performs better than many of the comparable methods, and the overall visual and parametric quality of the fusion results is also improved.**

Keywords—image fusion, multi-modal medical image fusion, SWT, wiener filter, method noise.

I. INTRODUCTION

The objective of image fusion is to generate a new image by amalgamating the attributes of many input images. The finished output, including all crucial data inside a single image, is more precise and instructive than any one source image. The twin objectives of image fusion, namely minimizing data volume and enhancing image quality for both human and machine observers, are well-established [1]. One application of image fusion is multi-sensor image fusion, which entails integrating relevant data from several source images into a single output image. Image fusion, a captivating domain within image processing, employs several methodologies to provide a cohesive image for analysis and diagnosis. Image fusion requires the completion of five essential steps: registration, decomposition, fusion rules, reconstruction, and evaluation procedures [2].

Various categories of image fusion are depending on the dataset, including multi-focus, multi-sensor, multi-scale, multi-spectral, and multi-temporal. A single input image that preserves all pertinent information may be generated by amalgamating several input images with differing focal points using multi-focus image fusion techniques. Medical diagnostic applications use multi-sensor methodologies for image fusion, amalgamating data from several sources to provide a more complete and intelligible image. This data may be imperceptible to the unaided human eye. It is feasible to accurately identify the location of abnormalities using data obtained from composite images [3]. The multi-scale image fusion technique effectively integrates Low Dynamic Range (LDR) images with differing exposures into a single, more informative LDR image. Multi-spectral image fusion techniques use data integration in the spectrum domain to exploit related information and spatiotemporal correlations, resulting in a coherent image. Information failure is reduced, and comprehensive information of all clinical parameters are obtained by multi-temporal image fusion techniques [4, 5]. Fig 1 illustrates an example of multi-modal medical image fusion.

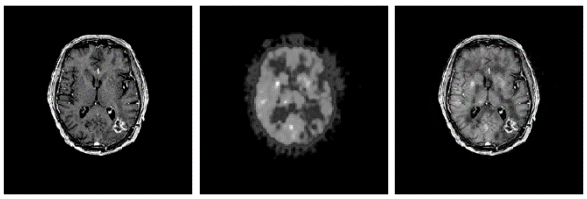


Fig 1. Example of multi-modal medical image fusion

Medical practitioners use many imaging modalities for disease diagnosis, including positron emission tomography (PET), T1-weighted and T2-weighted magnetic resonance images (MR-T1 and MR-T2), CT, magnetic resonance angiography (MRA), and MRI. These imaging technologies may provide an abundance of data on a patient's health, are cost-effective, and user-friendly. Numerous advantages exist, although articulating them remains a significant difficulty [6]. Even highly skilled radiologists may find it challenging to detect tiny nodules or differentiate between similar malignancies. Manual disease screening involves considerable time and effort. Image fusion has developed as a very promising domain of image processing due to its crucial importance in several applications, including medical diagnosis and image clarity. Multimodal Medical Image Fusion enhances medical image quality by integrating several images obtained from different modalities. The outcome is a composite image that surpasses the aggregate of its components [7].

This work seeks to develop a multimodal medical imaging technique that integrates SWT and Wiener filter with MRI and CT images, using several quality criteria to optimize the information content in the fused image. Subsequently, it will be contrasted with prior fusion techniques. Incorporating a

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Wiener filter with method noise as a post-processing step enhances the outcomes of this hybrid methodology by introducing a noise mitigation strategy.

After the introduction, Section 2 of the article details all the relevant theories that informed this study. The technique is described in Section 3, along with an algorithm and a flow diagram. The article is concluded in Section 5, after which Section 4 delves into the outcomes of fusion.

II. Related Theory

A. Image Fusion using Stationary Wavelet Transform

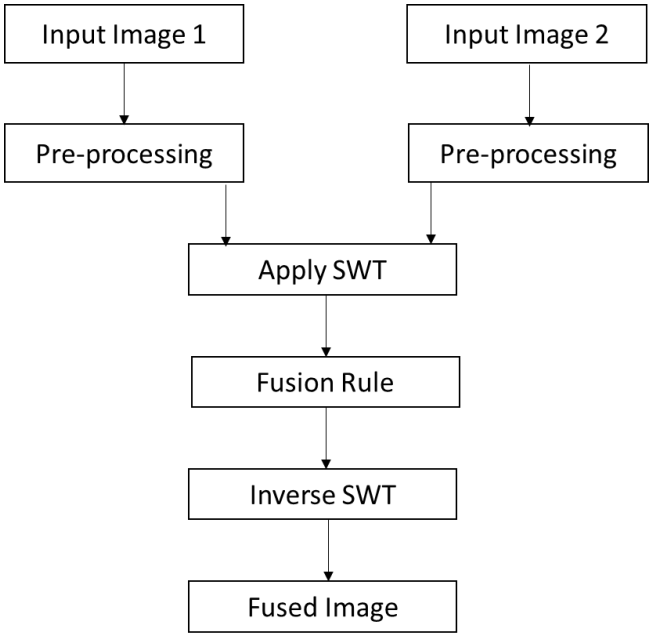


Fig 2. General image fusion process using SWT [8]

The general image fusion process using SWT method is shown in Fig 2. Here the image is divided into low frequency (approximation) and high frequency (details). There are four sub-signals at each level. The estimate displays the general pattern of the pixel values as well as the specifics in the form of the diagonal, vertical, and horizontal components. The initial step in this image fusion method is to improve the quality of the two input images, which are called Image 1 and Image 2. After that, one can take the improved images and put them into the wavelet domain, which separates them into four parts: the horizontal, diagonal, approximation, and the image itself [7].

Individual image features may be dissected in great depth in the wavelet domain to image decomposition. The next stage is to analyze the wavelet coefficients of the two images to filter out unnecessary or irrelevant features and optimize the extraction of important information. To save the most useful parts of both images, this is the stage when fusion rules may be implemented. The inverse wavelet transform is executed after the optimum information from the wavelet coefficients has been fused. By combining the data from the two input images, this operation creates a new image with improved quality and more detailed features than the originals.

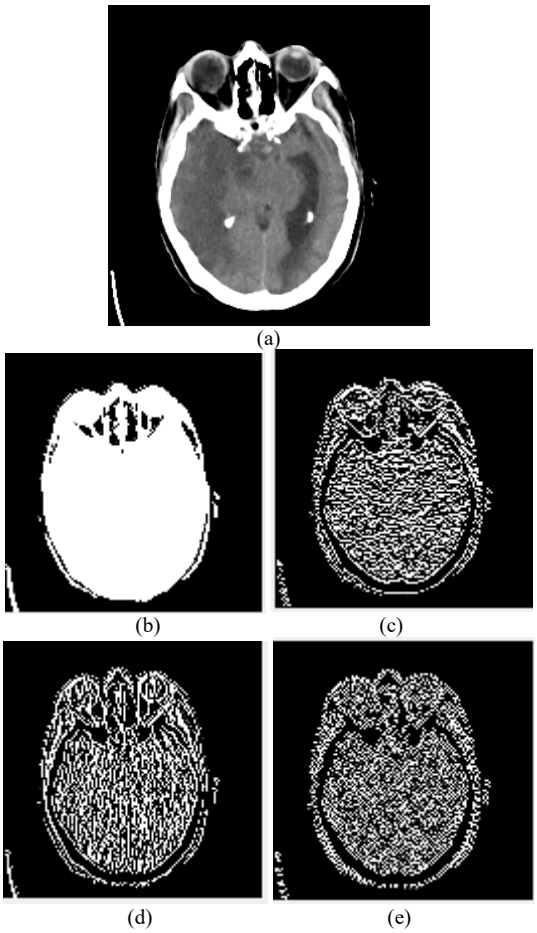


Fig 3. (a) Original image before to alteration (b–e) Decomposition of image into sub-signals

Above is a breakdown of the original CT image into its constituent parts: the average (approximate), the horizontal, the vertical, and the diagonal areas as shown in Fig 3. SWT is utilized in this method to integrate images from several imaging modalities, such as MRI and CT scans. The SWT may be used to separate the two images based on their frequency components. After the images have been broken down, their wavelet components are subjected to an efficient fusion algorithm. Applying this fusion algorithm to the data from each modality results in a final fused image that enhances diagnostic value by integrating the qualities of both imaging modalities.

B. Wiener Filter and Method Noise

When dealing with blurred or noise-damaged images, the Wiener filter is the stationary linear filter that maximizes mean squared error. One of the first approaches to reducing additive random noise in images was Wiener filtering. Due to their inherent limitations, Wiener filters cannot restore noise-damaged frequency components. Method noise is used in the post-processing step. Improving the image fusion process is possible by combining the wiener filter with method noise. Even further reduction of background noise and unwanted information is possible after adding method noise to the image's residuals using the wiener filter. The wiener filter is applied on the residual parts obtained from the subtracting fused image from input images. This has a dual purpose: first, it improves the overall appearance of the fused image while simultaneously enhancing its clarity and quality by removing artifacts [8].

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III. Proposed Methodology: M3IF-SWT

A. Proposed Flowchart: M3IF-SWT

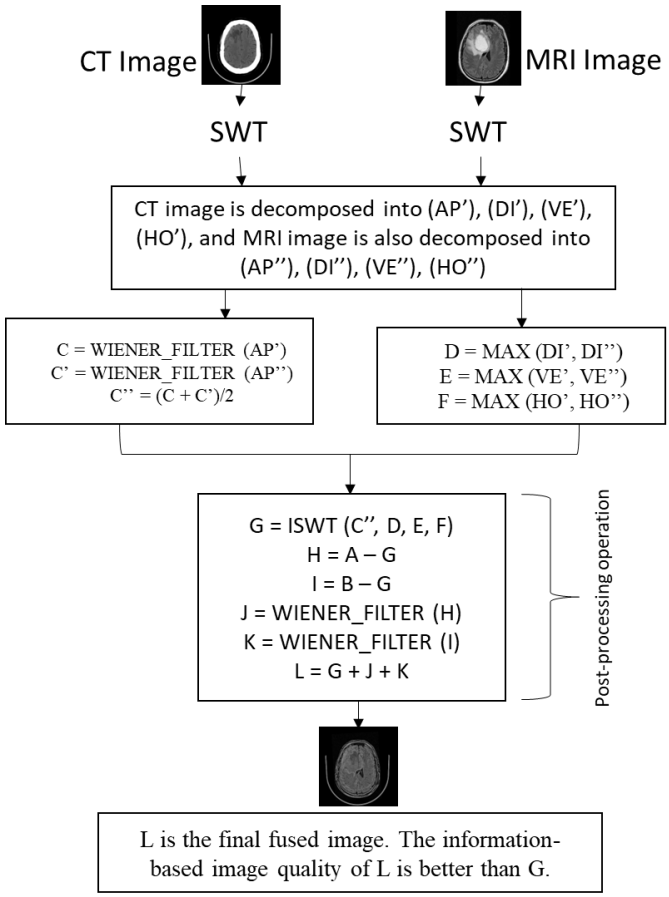


Fig 4. Flowchart of proposed methodology

B. Proposed Algorithm: M3IF-SWT

Step 1. Apply SWT on input image 1 (CT image), A

Step 2. Apply SWT on input image 2 (MRI image), B

Step 3. CT image is decomposed into approximate (AP'), diagonal (DI'), vertical (VE'), and horizontal components (HO').

Step 4. Similarly, MRI image is also decomposed into approximate (AP''), diagonal (DI''), vertical (VE''), and horizontal components (HO'').

Step 5. The approximate part of CT image is directed to wiener filter.

$$C = WIENER_FILTER\ (AP')$$

Step 6. The approximate part of MRI image is directed to wiener filter.

$$C' = WIENER_FILTER\ (AP'')$$

Step 7. Take average of C and C'

$$C'' = (C + C')/2$$

Step 8. The diagonal parts of CT image and MRI image are processed using maximum operation.

$$D = MAX\ (DI', DI'')$$

Step 9. The vertical parts of CT image and MRI image are processed using maximum operation.

$$E = MAX\ (VE', VE'')$$

Step 10. The horizontal parts of CT image and MRI image are processed using maximum operation.

$$F = MAX\ (HO', HO'')$$

Step 11. Step 7 averages out the information and filters it as well by deblurring, while Step 8-10 extract the best information out of it.

Step 12. Inverse DWT is applied to obtain the fused image as shown in below equation. G is the fused image.

$$G = IDWT\ (C'', D, E, F)$$

Step 13. The concept of method noise is used as post-processing operation to improve the results conventional image fusion method using DWT.

Step 14. H = A – G

Step 15. I = B – G

Step 16. Wiener Filter is applied on residual part H and I to remove the noise and improve the residual parts.

$$J = WIENER_FILTER\ (H)$$
$$K = WIENER_FILTER\ (I)$$

Step 17. L = G + J + K

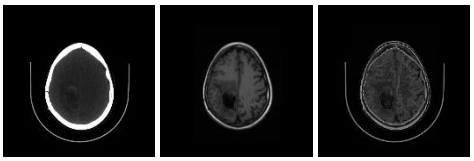
Step 18. L is the final fused image. The information-based image quality of L is better than G, which is obtained from traditional SWT method.

C. Significance of M3IF-SWT

The M3IF-SWT achieves improved outcomes through the application of method noise and the Wiener filter as a post-processing operation. The integration of various methodologies enhances the outcomes of the standard SWT method in image fusion. The initial blurring issue is addressed by directly implementing the Wiener filter on the approximate part. This indicates that the post-processing operation is applicable in various image enhancement and restoration models, owing to the effectiveness of the method noise concept. The working flowchart of M3IF-SWT is shown in Fig 4.

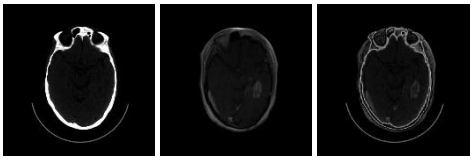
IV. Results And Discussion

Figures 5–8 illustrate the datasets of CT and MRI images [9 - 11]. Figures 5–8 show the visual quality outcomes of M3IF-SWT on various datasets. Zooming into the images reveals that the results are good, and that the post-processing procedure used is successful. The edges and corners are effectively preserved. Uniformity is maintained, and no information is lost. It is evident that the M3IF-SWT is an upgraded version of the traditional SWT approach.



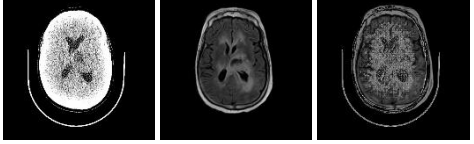
(a) (b) (c)

Fig 5. Fusion results of M3IF-SWT (a) CT image (b) MRI image (c) Fused image



(a) (b) (c)

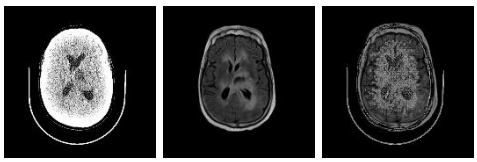
Fig 6. Fusion results of M3IF-SWT (a) CT image (b) MRI image (c) Fused image



(a) (b) (c)

Fig 7. Fusion results of M3IF-SWT (a) CT image (b) MRI image (c) Fused image

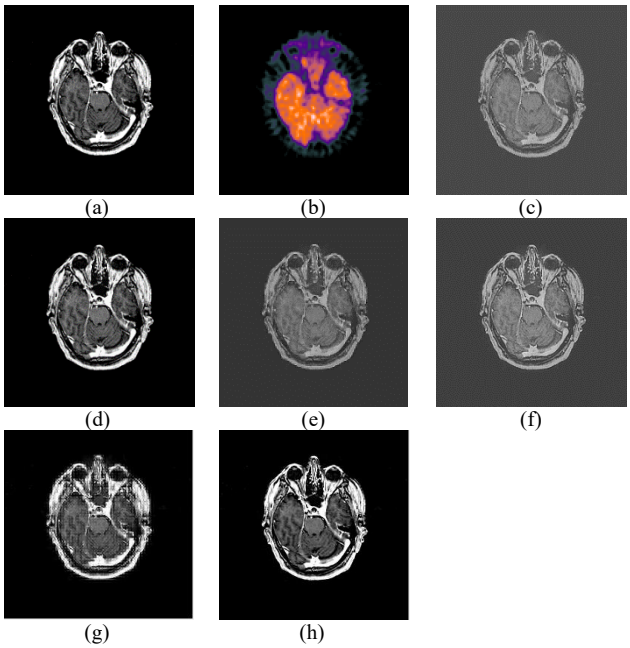
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(a) (b) (c)

Fig 8. Fusion results of M3IF-SWT (a) CT image (b) MRI image (c) Fused image

The results of M3IF-SWT are also compared with various standard methods as shown in Fig 9. The M3IF-SWT is compared with conventional methods like Modified Singular Value Decomposition (MSVD), DWT, SWT, Principal Componnet Analysis (PCA), and Wavelet Packet Decomposition (WPD). These results are comparatively better than these traditional methods.



(a) (b) (c) (d) (e) (f) (g) (h)

Fig 9. (a) MRI image (b) PET image (c) MSVD (d) PCA (e) DWT (f) SWT (g) WPD (h) M3IF-SWT

Upon thorough examination of the visual outcomes (Fig 9) and the parametric data (Figs 10, and 11), it can be concluded that the M3IF-SWT demonstrates superior performance relative to the other methods evaluated. While the comparative performance of M3IF-SWT does not significantly exceed expectations, the results are nonetheless evident, but still comparatively the results are better than conventional methods. It can be concluded that the application of smart post-processing techniques, specifically utilizing noise reduction and wiener filtering on the SWT, has yielded slightly improved outcomes.

There is no reference image for the comparison, therefore, objective method performance assessment metrics without a reference image are used for the overall comparison of M3IF-SWT with other methods. The metrics used are Functional Mutual Information (FMI), Fusion Symmetry (FS), Mean (M), Fusion Factor (FF), Standard Deviation (SD), Entropy (E), Fused Index (FI), and Sum of Correlation Difference (SCD). Except FS, all the metrics should be having higher values for the better results. Only FS should have lower value for the better results.

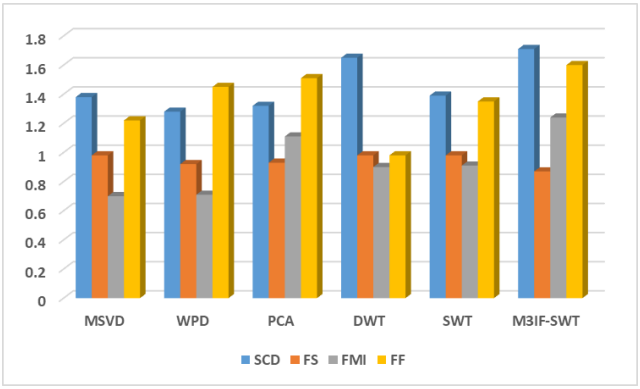


Fig 10. Graphical analysis based on the SCD, FS, FMI, and FF

The graphical representation of the data presented in Figs in 10 and 11. This facilitates a more comprehensive understanding parametric dat. The findings presented represent average outcomes derived from experiments conducted on over 80 datasets.

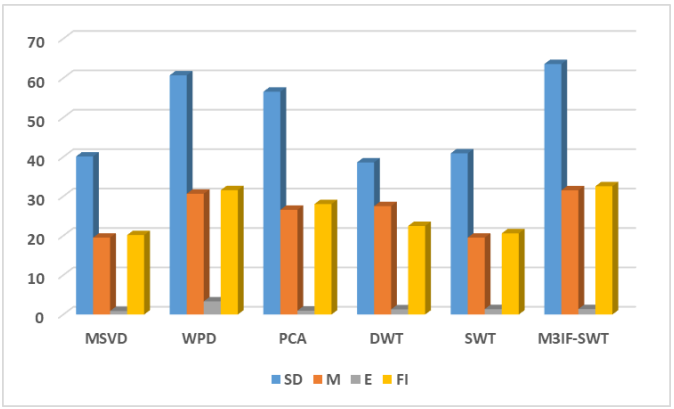


Fig 11. Graphical analysis based on the SD, M, E, and FI

V. Conclusion

The M3IF-SWT method presents a robust approach for the integration of CT and MRI images, along with various other medical modalities, in a fusion process. This methodology enhances the quality of fused images through the application of the SWT and method noise-based wiener filter as a post-processing operation. Following the execution of the SWT on both CT and MRI images, the resulting wavelet coefficients are combined utilizing a sophisticated fusion rule to extract the most optimal information. The findings related to fusion are enhanced through the application of the method noise concept, which operates in conjunction with the Wiener filter as a post-processing step. This method enhances both the visual quality and parametric performance of the combined images. The M3IF-SWT method demonstrates significantly improved diagnostic utility and clarity in comparison to traditional SWT methods and alternative approaches. This capability enhances clinical decision-making by generating fused images that exhibit greater accuracy and provide more informative insights.

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